

Network Scanning with N-map

Network Scanning : It is the process of discovering devices, open ports, and services on a network. It helps in identifying **security vulnerabilities**, **monitoring network activity**, and **detecting unauthorized devices**.

A network scan helps you map everything in the network and find security weaknesses.

N-map (Network Mapper) : It is a tool that helps you check what's happening on a network. It can:

- Find **devices** connected to a network.
- Check which **ports** (doors) are open on a device.
- See what **services** (programs) are running on those ports
- Guess the operating system (Windows, Linux, etc.)
- Check for security risks

Perform different types of scans:

1. **Basic scan** : `nmap <IP>` : shows the **port number, state and services** (if detected)

→ to know the IP address of your system run :

`ip a`

or

`hostname -I`

```
aditi@aditi-IdeaPad-3-15IAU7:~$ hostname -I
aditi@aditi-IdeaPad-3-15IAU7:~$ nmap 
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-11 21:19 IST
Nmap scan report for aditi-IdeaPad-3-15IAU7 ( )
Host is up (0.00012s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
25/tcp    open  smtp
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
```

21/tcp (FTP) - Transfers files, vulnerable if not secured.

25/tcp (SMTP) - Sends emails, can be abused for spamming.

80/tcp (HTTP) - Hosts a website, not secure.

443/tcp (HTTPS) - Secure web server with SSL/TLS encryption.

2. **Service detection** : `nmap -sV <IP>` : detects the **version of services running** on open ports and provides more details about the **software** used.

```
aditi@aditi-IdeaPad-3-15IAU7:~$ nmap -sV [REDACTED]
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-11 21:31 IST
Nmap scan report for aditi-IdeaPad-3-15IAU7 ([REDACTED])
Host is up (0.00012s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE  VERSION
21/tcp    open  ftp      ProFTPD
25/tcp    open  smtp     Postfix smtpd
80/tcp    open  http     Apache httpd 2.4.58 ((Unix) OpenSSL/1.1.1w PHP/8.2.12 mod_perl/2.0.12 Perl/v5.34.1)
443/tcp   open  ssl/http Apache httpd 2.4.58 ((Unix) OpenSSL/1.1.1w PHP/8.2.12 mod_perl/2.0.12 Perl/v5.34.1)
Service Info: Host: aditi-IdeaPad-3-15IAU7

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.77 seconds
aditi@aditi-IdeaPad-3-15IAU7:~$
```

Version detected : ProFTPD, Postfix smtpd, Apache httpd 2.4.58 (Unix) and Apache httpd 2.4.58 (Unix) with OpenSSL 1.1.1w, PHP 8.2.12, mod_perl 2.0.12, Perl v5.34.1

Software info & Components detected :

- Open SSL 1.1.1w (for secure communication)
- PHP 8.2.12 (web server supports PHP scripts)
- mod_perl 2.0.12 (Apache module that allows running Perl scripts)\
- Perl v5.34.1 (version)

note : A Perl script is a program written in the Perl programming language, which is used for text processing, automation, web development, and system administration.

3. **Aggressive scan**: `nmap -A <IP>` : This scan provides **detailed information** about **open ports, running services, SSL certificates, and potential security issues**.

It includes :

- OS Detection (-O)
- Version Detection (-sV)
- Traceroute (--traceroute) {Maps the network path to the target.}
- Script Scanning (--script=default) {check for vulnerabilities and misconfigurations.}

```
aditi@aditi-IdeaPad-3-15IAU7:~$ nmap -A [REDACTED]
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-02-11 22:07 IST
Nmap scan report for aditi-IdeaPad-3-15IAU7 ([REDACTED])
Host is up (0.00033s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD
25/tcp    open  smtp      Postfix smtpd
| ssl-cert: Subject: commonName=ubuntu
| Subject Alternative Name: DNS:ubuntu
| Not valid before: 2024-04-08T07:25:13
|_Not valid after: 2034-04-06T07:25:13
|_ssl-date: TLS randomness does not represent time
|_smtp-commands: aditi-IdeaPad-3-15IAU7, PIPELINING, SIZE 10240000, VRFY, ETRN,
STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN, SMTPUTF8, CHUNKING
80/tcp    open  http      Apache httpd 2.4.58 ((Unix) OpenSSL/1.1.1w PHP/8.2.12 mod
_perl/2.0.12 Perl/v5.34.1)
|_http-title: Site doesn't have a title (text/html).
| http-methods:
|_ Potentially risky methods: TRACE
|_http-server-header: Apache/2.4.58 (Unix) OpenSSL/1.1.1w PHP/8.2.12 mod_perl/2.
0.12 Perl/v5.34.1
443/tcp   open  ssl/http  Apache httpd 2.4.58 ((Unix) OpenSSL/1.1.1w PHP/8.2.12 mod
_perl/2.0.12 Perl/v5.34.1)
```

```
| http-methods:
|_ Potentially risky methods: TRACE
|_http-server-header: Apache/2.4.58 (Unix) OpenSSL/1.1.1w PHP/8.2.12 mod_perl/2.
0.12 Perl/v5.34.1
443/tcp   open  ssl/http  Apache httpd 2.4.58 ((Unix) OpenSSL/1.1.1w PHP/8.2.12 mod
_perl/2.0.12 Perl/v5.34.1)
| http-methods:
|_ Potentially risky methods: TRACE
|_http-title: Site doesn't have a title (text/html).
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=localhost/organizationName=Apache Friends/stateO
rProvinceName=Berlin/countryName=DE
| Not valid before: 2004-10-01T09:10:30
|_Not valid after: 2010-09-30T09:10:30
|_http-server-header: Apache/2.4.58 (Unix) OpenSSL/1.1.1w PHP/8.2.12 mod_perl/2.
0.12 Perl/v5.34.1
| tls-alpn:
|_ http/1.1
Service Info: Host: aditi-IdeaPad-3-15IAU7

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.15 seconds
aditi@aditi-IdeaPad-3-15IAU7:~$
```

Basic Info :

- Host is up (0.00033s latency)
- Not shown: 996 closed ports (conn-refused)
- 4 open ports detected (21, 25, 80, 443)

Port Wise Breakdown :

PORT	STATE	SERVICE	VERSION	Explanation
21/tcp	open	FTP (File Transfer Protocol)	ProFTPD	FTP service is running, used for transferring files. If not secured, it may allow unauthorized file access.
25/tcp	open	SMTP (Simple Mail Transfer Protocol)	Postfix smtpd	Used for sending emails. Open SMTP ports may allow spam or email spoofing.
80/tcp	open	HTTP (Web Server)	Apache 2.4.58 (Unix)	A website is running, but it does not have a title.
443/tcp	open	HTTPS (Secure Web Server)	Apache 2.4.58 (Unix) with OpenSSL and PHP	A secure website is running over SSL encryption.

SMTP SSL Certificate :

Common name : Ubuntu
Valid from: 2024-04-08
Expires on: 2034-04-06

HTTPS SSL Certificate:

Common Name (CN): localhost

- Organization: Apache Friends
- Issued in: Berlin, DE (Germany)
- Valid from: 2004-10-01
- Expired on: 2010-09-30

Problem: The HTTPS certificate is expired, which makes it insecure. Attackers can perform MITM (Man-in-the-Middle) attacks.

⚠ Security Issues Detected

1 🚫 HTTP Server Misconfiguration

- `http-methods: TRACE` method is enabled, which can be used for **Cross-Site Tracing (XST)** attacks.
- **Risk:** Attackers can steal cookies or session information.
- **Fix:** Disable `TRACE` in Apache configuration.

2 🚫 Expired SSL Certificate on HTTPS (Port 443)

- **Risk:** Users visiting the website may get security warnings. Attackers can intercept encrypted communication.
- **Fix:** Renew the SSL certificate.

3 🚫 SMTP Service Running on Port 25

- **Risk:** If misconfigured, it may allow **spam emails or email spoofing**.
- **Fix:** Use proper authentication and encryption (TLS/SSL).

4 🚫 TLS Randomness Warning (`_ssl-date: TLS randomness does not represent time`)

- **Risk:** Indicates an issue with how SSL/TLS randomness is generated, which could weaken encryption.
- **Fix:** Update OpenSSL and verify the system clock.

Q) What services are running? Any vulnerabilities detected?

Ans : Services running : 1) FTP 2) SMTP 3) HTTP 4) HTTPS

Vulnerabilities :

1. FTP (Port 21 – ProFTPD) : Risk of unauthorized access if anonymous login is enabled.
2. SMTP (Port 25 – Postfix smtpd) : Possible email spamming if open relay is enabled.
3. HTTP (Port 80 – Apache 2.4.58)
 - Outdated Apache version
 - `TRACE` method enabled (can be used for **cross-site tracing (XST) attacks**).
4. HTTPS (Port 443 - Apache 2.4.58 SSL/TLS):

- Expired SSL certificate
- TLS configuration detected

Remember :

- Scanning a network means checking all connected devices and their open ports.
- It helps find security risks and unknown devices.
- Nmap is a powerful tool to scan network easily.